An entropy action across wirings: Bianconi's discrete entropic action evaluated on networks that rewire

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Bianconi's information-theoretic action for network geometry is defined on a cell complex with fixed wiring, and its author names the dynamics of the wiring as open. We evaluate that action, unchanged, on networks whose wiring does change: four-regular bipartite graphs rewired under a family of square-counting energies that contains combinatorial quantum gravity as one member, with every square taken as a 2-cell. (i) The construction passes the identities the source states. (ii) In the action's vacuum the metric is the same on every cell, so the action is one number per cell and, across wirings with the same number of vertices, a function of the number of squares alone; it orders wirings exactly as the square-counting energy without its local term does. (iii) With the induced metric $\mathbf{I} + c_0\boldsymbol{\mathcal{L}}$, which depends on the wiring, evaluated at the identity metric, the action on 445 saved wirings of 64 vertices falls with every square and rises with every surplus square, as that energy does, with an effective local coefficient $\lambda_{\text{eff}} = 0.001$ to 0.55 for $c_0 = 0.01$ to 100. That is below 1, the range in which curled arrangements lie lower than the flat torus. The metric that solves the action's own equation for $c_0 > 0$ is not computed, and whether a comparison across wirings is the intended one is put as a question to its author. Nothing here is a statement about gravity.

I. INTRODUCTION

Ref. [1] proposes an action for a network geometry coupled to matter and gauge fields: the quantum relative entropy between a metric on the cells of a complex and the metric that matter induces on them. It is the discrete form of "gravity from entropy" [2]. The complex is held fixed. The action "depend[s] also on the topology of the higher-order networks," and the paper leaves "the possible implied dynamics of the network topology to future works" (its Sec. 3.1 and its conclusions).

The companion paper [3] studies networks that do rewire, under an energy that counts squares. We ask one narrow question: read in the most direct way, what does the action of Ref. [1] say about arrangements with the same vertices and different wiring?

Three kinds of statement are kept apart. *Sourced*: taken from Ref. [1], with its equation numbers. *Exact*: linear algebra on named graphs, reproducible from the repository. *Ours*: a reading or a fit, unreviewed. The reading of the action across wirings is ours throughout.

II. SETTING

The networks. Four-regular bipartite graphs on N vertices in which no two vertices share more than two neighbors. With S the number of 4-cycles (squares), S_e the squares on link e and $X = \sum_e (S_e - 2)_+$ the surplus squares, the energy is

$$H = 16(N - S) + 4\lambda X. \tag{1}$$

Only $\lambda = 1$ is combinatorial quantum gravity (CQG) [4, 5]; other values are our family around it. Three arrangements of the same vertices are the rungs of a ladder [3]: the flat torus ($S = N$, $X = 0$), the curled torus, four steps around in one direction ($S = 5N/4$, $X = N$), and the 4-cube, curled both ways ($S = 3N/2$, $X = 2N$). At $\lambda = 1$ the three have equal energy; below 1 the curled ones lie lower; above 1 the flat torus is lowest.

The complex (ours). Vertices, the $2N$ links, and every square as a 2-cell: $\mathcal{N} = 3N + S$ cells. The square count used here agrees with the simulation code's and with $(\mathrm{Tr}A^4 - 28N)/8$, where $28N = N(2 \cdot 4^2 - 4)$ counts the closed 4-walks that trace no square.

The action (sourced). With boundary matrices $\mathbf{B}_{[1]}, \mathbf{B}_{[2]}$ and a block-diagonal metric $\boldsymbol{\mathcal{G}}$ on the cells, Ref. [1] defines the weighted exterior derivative and the Dirac operator (its Eq. 31),

$$\mathbf{d} = \boldsymbol{\mathcal{G}}^{-1/2}\mathbf{B}^\dagger\boldsymbol{\mathcal{G}}^{1/2}, \qquad \mathbf{D} = \mathbf{d} + \mathbf{d}^\dagger,$$

whose square is the Gauss–Bonnet Laplacian $\boldsymbol{\mathcal{L}} = \mathbf{L}_{[0]} \oplus \mathbf{L}_{[1]} \oplus \mathbf{L}_{[2]}$ (its Eqs. 35–40). The action, which Ref. [1] (its Eq. 41) writes as

$$\mathcal{S}_+ = \sigma\,\mathrm{Tr}\ln\boldsymbol{\mathcal{G}} + \mathrm{Tr}\,\boldsymbol{\mathcal{G}}\big(\ln\boldsymbol{\mathcal{G}} - \ln\mathbf{G}\big) - \mathrm{Tr}\,\boldsymbol{\mathcal{G}},$$

compares $\boldsymbol{\mathcal{G}}$ with the induced metric $\mathbf{G}$. With no matter and no gauge field, $\mathbf{G} = \mathbf{I} + c_0\boldsymbol{\mathcal{L}}$ (its Eq. 52). Its vacuum is $\mathbf{G} = \mathbf{I}$, where the metric obeys $-\boldsymbol{\mathcal{G}}\ln\boldsymbol{\mathcal{G}} = \sigma\mathbf{I}$ (its Eq. 65).

III. THE CONSTRUCTION, CHECKED

Ref. [1] reports no number to reproduce, so what is reproduced is the construction. On every graph used here (exact): $\mathbf{B}_{[1]}\mathbf{B}_{[2]} = 0$; at the identity metric $\mathbf{D}^2$ equals $\boldsymbol{\mathcal{L}}$ block by block; $\mathbf{D}$ anticommutes with γ_0 (+1 on vertices

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and squares, -1 on links; its Eq. 34); and the numbers of zero modes of $\mathbf{L}_{[0]}, \mathbf{L}_{[1]}, \mathbf{L}_{[2]}$ are the Betti numbers of the complex: $(1, 2, 1)$ for the flat torus. For the curled torus they are $(1, 1, N/4)$: each ring of four is itself a square, a filled disk, so only the long loop survives, and two neighboring disks with the four squares between them close a cavity. A 4-cube gives $(1, 0, 7)$ (Euler characteristic $16 - 32 + 24 = 8$).

IV. THE VACUUM COUNTS CELLS

Exact, given the reading. With $\mathbf{G} = \mathbf{I}$ and the same root g of $-g \ln g = \sigma$ on every cell, Ref. [1]'s Eq. 41 becomes, using $g \ln g = -\sigma$,

$$\mathcal{S}_+ = \mathcal{N}\, s(\sigma), \qquad s(\sigma) = \sigma \ln g - \sigma - g. \tag{2}$$

The wiring enters only through the number of cells. Since $\mathcal{N} = 3N + S$ and $S = N - H_{\lambda=0}/16$,

$$\mathcal{S}_+ = 4N\, s - \frac{s}{16}\, H_{\lambda=0}, \tag{3}$$

and $s < 0$ (for example $g = 0.894194$, $s = -1.005377$ at $\sigma = 0.1$). At fixed N the vacuum action is therefore an increasing function of the square-counting energy with no local term: the two order every wiring alike. That energy is the one under which the network is reported to break into isolated hypercubes [4, 5]. Two cautions. For $\sigma < 1/e$ the source's Eq. 65 has two roots, and a metric may take either on each eigenvector; we take the root that tends to 1 as $\sigma \to 0$ everywhere. And "more negative" is not "favored" until a weight over wirings is given, which the source does not give.

V. AT THE IDENTITY METRIC

For $c_0 > 0$ the induced metric depends on the wiring through the Laplacian. At $\boldsymbol{\mathcal{G}} = \mathbf{I}$, where $\mathbf{d} = \mathbf{B}^\dagger$, the first term of the action vanishes and $\mathrm{Tr} \ln \mathbf{G} = \sum_i \ln(1 + c_0 \mu_i)$, so

$$\mathcal{S}_+ = -\sum_i \ln(1 + c_0 \mu_i) - \mathcal{N}, \tag{4}$$

with μ_i the eigenvalues of the three Hodge Laplacians. This is the action at one uniform metric, not at a solution of its equations (ours).

The ladder (exact). Table I: the more squares, the more negative the action. The order, 4-cubes below curled torus below flat torus, holds at every c_0 tried from 0.01 to 1000.

A twin (exact). Re-gluing two neighboring rings of the curled torus at opposite points gives a graph that is not isomorphic to it and has a different spectrum (eigenvalues differ by up to 0.39), with the same cells. The two agree in the first six moments of each Laplacian and in Eq. (4) to nine figures: they differ only through closed walks that go all the way round the long direction. At this size the action at the identity metric does not see the global twist.

TABLE I. Four arrangements of 64 vertices. Zero modes are those of $\mathbf{L}_{[0]}, \mathbf{L}_{[1]}, \mathbf{L}_{[2]}$. Vacuum: $\sigma = 0.1$. Exact.

	S	zero modes	vacuum	Eq. (4)	
			$\mathcal{S}_+$	$c_0 = 0.1$	$c_0 = 1$
flat, 8×8	64	1, 2, 1	-257.38	-339.46	-642.03
curled, 16×4	80	1, 1, 16	-273.46	-364.76	-688.13
curled, re-glued	80	1, 1, 16	-273.46	-364.76	-688.13
four 4-cubes	96	4, 0, 28	-289.55	-390.07	-734.78

TABLE II. Fit of Eq. (4) to $a + bS + cX$ over 445 wirings of 64 vertices: end states saved by the pre-registered runs of Ref. [3] (at most 25 of each kind (S, X); 42 kinds, $61 \le S \le 96$) and the three rungs. $\lambda_{\text{eff}} = -4c/b$. Exact for each wiring; the fit is exploratory.

c_0	b	c	λ_{eff}	rms left over
0.01	-1.078	0.0002	0.001	0.000
0.1	-1.645	0.015	0.037	0.014
1	-3.980	0.250	0.25	0.36
10	-7.863	0.891	0.45	2.4
100	-11.91	1.62	0.55	6.2

Why a penalty appears (ours). Expanding $\ln(1 + x) = x - x^2/2 + \ldots$ in Eq. (4) gives, for small c_0,

$$\mathcal{S}_+ = -\mathcal{N} - c_0\, \mathrm{Tr}\boldsymbol{\mathcal{L}} + \tfrac{1}{2} c_0^2\, \mathrm{Tr}\boldsymbol{\mathcal{L}}^2 - \ldots, \tag{5}$$

and $\mathrm{Tr}\boldsymbol{\mathcal{L}} = 8N + 8S$ (each vertex has four links, each link two ends and S_e squares, each square four sides): to first order the action counts squares and nothing else. At second order pairs of squares that share a link enter with the opposite sign. Crowded links are penalized without that being put in, and the surplus term of Eq. (1) is one measure of crowding.

Many wirings (exact per wiring; the fit is ours). Table II: every square lowers the action and every surplus square raises it. Since $bS + cX = (-b/16)(-16S + 4\lambda_{\text{eff}} X)$ with $-b > 0$, the action orders these wirings as H of Eq. (1) would with $\lambda = \lambda_{\text{eff}}$, which grows with c_0 and stays below 1 (0.034, 0.20 and 0.28 at $c_0 = 0.1$, 1 and 10 if the curled torus and the 4-cubes are left out). The linear form fits well at small c_0 and less well as c_0 grows.

VI. BESIDE COMBINATORIAL QUANTUM GRAVITY

Read at the identity metric, the action of Ref. [1] places these networks at $\lambda < 1$ in the family of Eq. (1), the side where curling pays, for every c_0 tried. It contains the ingredient that saturates the reward for squares, but too weakly to make the flat torus the lowest.

What this does not show, besides whether the comparison across wirings is the one intended (Sec. VII). Anything at a metric that solves the equation of motion $\sigma\boldsymbol{\mathcal{G}}^{-1} + \ln \boldsymbol{\mathcal{G}} = \boldsymbol{\mathcal{T}}$ (Ref. [1], its Eq. 57) for $c_0 > 0$, where the metric will differ from cell to cell and from wiring to wiring. Anything with matter or gauge fields. Anything about the continuum theory [2]. Anything about gravity.

VII. THE QUESTION

Is $\mathcal{S}_+$ compared across wirings at a fixed number of vertices the comparison meant by dynamics of the topology? If it is, must the metric be solved for each wiring first, or is something else meant to keep the action from rewarding more cells? The next steps follow from the answer: the metric equation solved on the ladder, where symmetry leaves a few unknowns per rung; the same on a flat sheet holding one of the leftover knots of the companion work; and six links per vertex.

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